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Thermodynamic modelling and optimization of the Al–Ce–Nd system

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Abstract

Using the Calphad approach a thermodynamic modelling and optimization of the Al–Ce–Nd systems has been carried out on the basis of the experimental investigation reported in a separate paper in this issue (Calphad (this issue)). A re-optimization of the Al–Nd binary subsystem was also necessary. The compound energy formalism has been used to describe the thermodynamic functions of the phases present in the systems. Optimization results, in good agreement with the experimental information, are presented and discussed.
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1. Introduction

It is well known that the addition of rare earths (R in the following) to Al-based alloys enhances their mechanical and corrosion-resistant properties [6, 8]. Several Al–R and Al–R–T alloys (with T = transition metal) are also interesting glass-forming systems, as discussed in different recent papers [7, 11, 14, 16].

Moreover it is also well known that, among the different alloy systems formed by the R metals combined with a specific partner, smooth and regular variations of properties and characteristics may be observed as a function of the R atomic number. Such systematic behaviour may be used as a *chemical* criterion in order to check the reliability of both experimental and computational results concerning, in particular, the constitutional properties of these systems.

Finally, it is recognized that, in order to improve the reliability and the predictive ability of the thermodynamic databases developed according to the Calphad approach, it is good practice to check the results obtained for lower order systems with the calculation and optimization of higher order systems.

In this paper the optimization of the Al–Ce–Nd ternary system is reported. It has been carried out on the basis of the experimental results obtained in our laboratory and reported in a separate paper in this issue [15]. The present optimization results will be discussed with particular reference to the recent optimization of the Al–Ce and Al–Nd binary subsystems [12].

2. Literature information

A summary of the literature data concerning the Al–Ce and Al–Nd binary systems is included in the recent optimization carried out by [12]. The ternary system has been investigated by [15] in the 25 to 67 at% Al composition range. In the same paper literature data concerning the constitutional properties of the three boundary binary systems are also reported.

3. Thermodynamic modelling

Thermodynamic models already introduced and described in [12] for the binary Al–Ce and Al–Nd phases have been maintained here. In addition, the mixing between Ce and Nd in the R-rich sublattice of the different solid phases has been considered. This is in agreement with the experimentally determined [15] ternary phase equilibria. In fact, isostructural phases in the two binary Al–Ce and Al–Nd subsystems form continuous solid solutions in the ternary phase, and phases with different structure in the binaries show quite large line-shaped solid solubilities, clearly due to substitution between Ce and Nd. Also from a crystallographic point of view it is reasonable to expect the atoms of the two light R to be able to substitute (at least partially) each other in the crystal lattice.

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Table 1
Names, crystallographic structures and sublattice models of the solid phases. Predominant species (if any) are underlined

Phase names		Pearson symbol-structure type	Names used in the Appendix	Sublattice model
Al–Ce	Al–Nd			
(Al), (γCe)	(Al)	cF4–Cu	A1	($\underline{\text{Al}}$, $\underline{\text{Ce}}$, Nd)
(δCe)	(βNd)	cI2–W	A2	(Al, $\underline{\text{Ce}}$, $\underline{\text{Nd}}$)
–	(αNd)	hP4–αLa	DHCP	(Al, Ce, $\underline{\text{Nd}}$)
$CeAl_2$	$NdAl_2$	cF24–$MgCu_2$	C15	(Al, $\underline{\text{Ce}}$, $\underline{\text{Nd}}$)$_1$($\underline{\text{Al}}$, Ce, Nd)$_2$
$CeAl_3$	$NdAl_3$	hP8–Ni_3Sn	NI3SN	(Al, Ce, Nd)$_1$(Al, Ce, Nd)$_3$
αCe_3Al	Nd_3Al			
CeAl	–	oC16–CeAl	CEAL	(Ce, Nd)$_1$(Al)$_1$
–	NdAl	oP16–ErAl	ERAL	(Ce, Nd)$_1$(Al)$_1$
–	Nd_2Al	oP12–Co_2Si	CO2SI	(Ce, Nd)$_2$(Al)$_1$
αCe_3Al_{11}	αNd_3Al_{11}	oI28–αLa_3Al_{11}	LA3AL11	(Ce, Nd)$_3$(Al)$_{11}$
βCe_3Al_{11}	βNd_3Al_{11}	tI10–Al_4Ba	AL4BA	(Ce, Nd)$_3$(Al)$_{11}$
βCe_3Al	–	cP4–$AuCu_3$	AUCU3	(Ce, Nd)$_3$(Al)$_1$

The list of the solid phases stable in the system, their crystal structures and the sublattice models here adopted are summarized in Table 1 and briefly introduced in the following.

3.1. Elements

The Gibbs energy of the pure elements is referred to as the enthalpy of the element in the standard element reference (SER) state. The Gibbs energy functions of the pure elements have been taken from the PURE database [5].

3.2. Phases

The Gibbs energy of the various phases is expressed according to the compound energy formalism [13]. According to this formalism, the different atoms forming the phase occupy one or more sublattices and thermodynamic quantities are expressed as a function of the site fractions. The model is suitable for the description of either stoichiometric compounds and of both disordered and ordered solutions.

In general the total Gibbs energy of a phase φ is given by the sum of three contributions:

$$G^{\varphi} = {}^{\text{ref}}G^{\varphi} + {}^{\text{id}}G^{\varphi} + {}^{\text{ex}}G^{\varphi}$$

where the three terms assume different forms according to the kind of sublattice model adopted.

3.3. Stoichiometric compounds

Stoichiometric compounds are represented with as many sublattices as the number of component elements, with only one atom type in each sublattice. In the binary case the model is $(\text{A})_u(\text{B})_v$ and the three contributions to the Gibbs energy are:

$$\begin{aligned} {}^{\text{ref}}G^{\varphi} &= u/(u+v)G_{\text{A}}^{\text{SER}} + v/(u+v)G_{\text{B}}^{\text{SER}} + L_{\text{A:B}}^{\varphi} \qquad (1)\\ {}^{\text{id}}G^{\varphi} &= 0 \\ {}^{\text{ex}}G^{\varphi} &= 0 \end{aligned}$$

where $L_{\text{A:B}}^{\varphi} = \Delta_f G(\varphi)$ is the Gibbs energy of formation of the phase referred to the component elements in the SER state. This model was adopted for most intermediate phases in the A–Ce and Al–Nd systems. In all cases the L interaction functions were assumed to be:

$$L = a + bT \qquad (2)$$

where a and b are temperature- and composition-independent parameters to be optimized.

3.4. Liquid and solid substitutional solutions

Substitutional solutions are represented by only one sublattice where all the atoms mix together. In the binary case (A, B) the three contributions to the Gibbs energy are:

$$\begin{aligned} {}^{\text{ref}}G^{\varphi} &= x_{\text{A}}G_{\text{A}}^{\varphi} + x_{\text{B}}G_{\text{B}}^{\varphi} \\ {}^{\text{id}}G^{\varphi} &= RT[x_{\text{A}}\ln(x_{\text{A}}) + x_{\text{B}}\ln(x_{\text{B}})] \\ {}^{\text{ex}}G^{\varphi} &= x_{\text{A}}x_{\text{B}}\sum_{\nu} {}^{\nu}L_{\text{A,B}}^{\varphi}(x_{\text{A}} - x_{\text{B}})^{\nu} \qquad (3) \end{aligned}$$

where G_{A}^{φ} and G_{B}^{φ} are the Gibbs energies of the pure elements in the φ structure and the L functions have the form indicated in Eq. (2). This model was adopted for the liquid and for the fcc, bcc and dhcp solid solutions.

3.5. Ordered solid solutions

Ordered solid solutions are represented by a two-sublattice model where each sublattice is preferentially occupied by a specific element but other elements are allowed to mix in the same sublattice. A binary phase of this kind may be represented by the model $(\underline{\text{A}}, \text{B})_u(\text{A}, \underline{\text{B}})_v$, the first sublattice being mainly occupied by A and the second by B. If y'_{A}, y''_{A}, y'_{B} and y''_{B} are the site fractions of the elements A and B in the first ($'$) and second ($''$) sublattice respectively, the fully ordered state corresponds to the conditions $y'_{\text{A}} = 1$, $y''_{\text{A}} = 0$, $y'_{\text{B}} = 0$, $y''_{\text{B}} = 1$; the fully disordered state corresponds to the condition $x_{\text{B}} = y'_{\text{B}} = y''_{\text{B}}$

(same composition in each sublattice and in the overall phase). The contributions to the Gibbs energy are:

$$
\begin{aligned}
{}^{\mathrm{ref}}G^{\varphi} &= y'_{\mathrm{A}}y''_{\mathrm{A}}G^{\varphi}_{\mathrm{A:A}} + y'_{\mathrm{A}}y''_{\mathrm{B}}G^{\varphi}_{\mathrm{A:B}} + y'_{\mathrm{B}}y''_{\mathrm{A}}G^{\varphi}_{\mathrm{B:A}} \\
&\quad + y'_{\mathrm{B}}y''_{\mathrm{B}}G^{\varphi}_{\mathrm{B:B}} \\
{}^{\mathrm{id}}G^{\varphi} &= RT\{u/(u+v)[y'_{\mathrm{A}}\ln(y'_{\mathrm{A}}) + y'_{\mathrm{B}}\ln(y'_{\mathrm{B}})] \\
&\quad + v/(u+v)[y''_{\mathrm{A}}\ln(y''_{\mathrm{A}}) + y''_{\mathrm{B}}\ln(y''_{\mathrm{B}})]\} \\
{}^{\mathrm{ex}}G^{\varphi} &= y'_{\mathrm{A}}y'_{\mathrm{B}}y''_{\mathrm{A}}\sum_{\nu}{}^{\nu}L^{\varphi}_{\mathrm{A,B:A}}(y'_{\mathrm{A}} - y'_{\mathrm{B}})^{\nu} \\
&\quad + y'_{\mathrm{A}}y'_{\mathrm{B}}y''_{\mathrm{B}}\sum_{\nu}{}^{\nu}L^{\varphi}_{\mathrm{A,B:B}}(y'_{\mathrm{A}} - y'_{\mathrm{B}})^{\nu} \\
&\quad + y'_{\mathrm{A}}y''_{\mathrm{A}}y''_{\mathrm{B}}\sum_{\nu}{}^{\nu}L^{\varphi}_{\mathrm{A:A,B}}(y''_{\mathrm{A}} - y''_{\mathrm{B}})^{\nu} \\
&\quad + y'_{\mathrm{B}}y''_{\mathrm{A}}y''_{\mathrm{B}}\sum_{\nu}{}^{\nu}L^{\varphi}_{\mathrm{B:A,B}}(y''_{\mathrm{A}} - y''_{\mathrm{B}})^{\nu} \\
&\quad + y'_{\mathrm{A}}y'_{\mathrm{B}}y''_{\mathrm{A}}y''_{\mathrm{B}}\sum_{\nu}{}^{\nu}L^{\varphi}_{\mathrm{A,B:A,B}}(y'_{\mathrm{A}} - y'_{\mathrm{B}})^{\nu} \\
&\quad \times (y''_{\mathrm{A}} - y''_{\mathrm{B}})^{\nu}. \qquad (4)
\end{aligned}
$$

It may be emphasized that when more atoms mix in a sublattice (see Eq. (3)) ${}^{\mathrm{ref}}G^{\varphi}$ requires the Gibbs energy of the pure elements in the φ phase to be known even if it is not a stable phase for the element. Similarly, when more sublattices are used and more atoms occupy each sublattice, terms of the type $G^{\varphi}_{\mathrm{A:B}}$ appear in ${}^{\mathrm{ref}}G^{\varphi}$ (see (4)). They correspond to the Gibbs energies of formation of hypothetical stoichiometric compounds metastable in the system. Nevertheless they have to be evaluated in some way, together with the ${}^{\nu}L^{\varphi}$ parameters (which still have the form shown in Eq. (2)).

This model was adopted here for the $MgCu_2$ type phase present in the system as a line compound included between the binary compositions $CeAl_2$ and $NdAl_2$. This was also used for the Ni_3Sn-type phase which is stable in the ternary system at different compositions corresponding to different site occupations of the same sublattices, as reported in Table 1.

4. Optimization procedure

According to the literature, the Ce–Nd binary equilibria have not been studied experimentally. On the other hand, considering the close similarities linking the two light rare earths, it is reasonable to assume ideal mixing between them in the liquid and solid phases. This is also supported by the results obtained in the optimization of a number of binary systems of adjacent or close rare earths (Tb–Ho, Tb–Er, Dy–Ho, Dy–Er, Ho–Er) carried out by [9]: the experimental phase diagrams were well reproduced by assuming ideal mixing both in the liquid and solid solutions. Accordingly no Ce–Nd binary interaction parameters were optimized and ideal mixing was assumed.

For the Al–Ce and Al–Nd systems the optimization results by [12] were initially adopted and used for a first calculation of the ternary system. In order to reproduce the line solubilities of the solid phases in the 500 °C isothermal section, binary interaction parameters corresponding to the metastable end-members of the binary R–Al phases were introduced. However, a satisfactory reproduction of experimental invariant phase compositions and tie-line orientations in selected areas of the phase diagram was impossible, even by adding ternary interaction parameters. Such impossibility was related to the difference between the optimized parameters of corresponding phases in the Al–Ce and Al–Nd subsystems. In some cases, it was too large and incompatible with the ternary equilibria.

As a consequence it was necessary to re-assess one of the two Al–R binary systems. Al–Nd was chosen, considering the relatively large entropy values resulting from the previous optimization. Parameters related to the entropy of formation or mixing of all the Al–Nd phases were then re-optimized. After the re-optimization the difference between the old and new version of the calculated phase diagram was well included within the experimental incertitude, and a satisfactory reproduction of the ternary isothermal equilibria was easily obtained.

Finally, in order to get an at least approximate evaluation of the ternary interaction parameters for the liquid phase, a few DTA experiments were carried out on selected samples at key compositions (see [15]) and their results used in the optimization.

5. Results and discussion

The optimization has been carried out by means of the Thermo-Calc software package [4]. Present results are reported in the Appendix, where the parameters calculated in this work are underlined.

5.1. The Ce–Nd binary system

The Ce–Nd phase diagram, as calculated in this work, is presented in Fig. 1. It may be compared with the hypothetical phase diagram reported in [10].

5.2. The Al–Ce binary system

The Al–Ce phase diagram has been taken from [12] and here reproduced in Fig. 2 for the reader's convenience. The negligible discrepancies with respect to the figure in [12] are due to the different model adopted here for the Ni_3Sn type phase.

5.3. Re-optimization of the Al–Nd binary system

The Al–Nd phase diagram, as re-optimized in this work, is presented in Fig. 3. It may be compared with the phase diagram computed in [12]: the differences are small and included in the uncertainties which can be assigned to the phase equilibria in this system. Thermodynamic functions

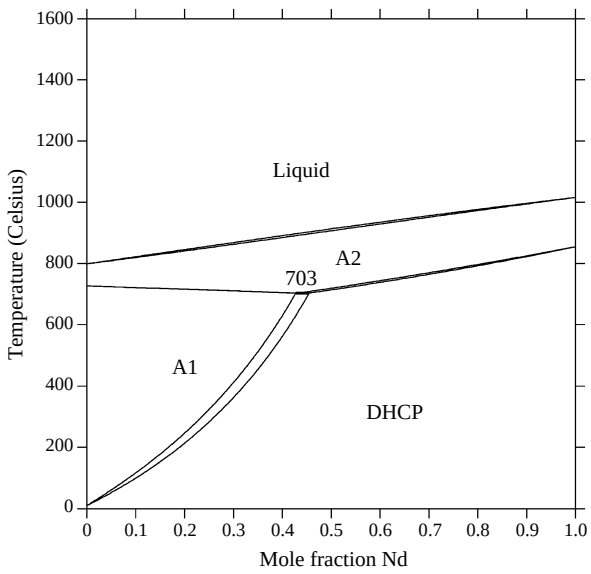

Fig. 1. The Ce–Nd system calculated assuming ideal mixing in all the solid and liquid phases.

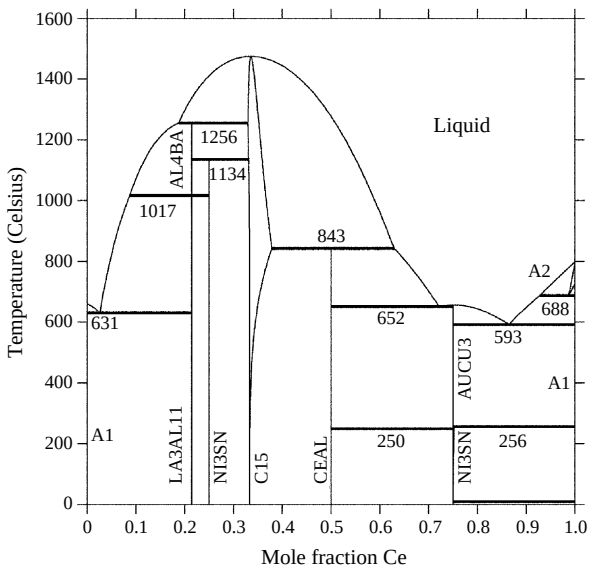

Fig. 2. The calculated Al–Ce system (according to [12]) .

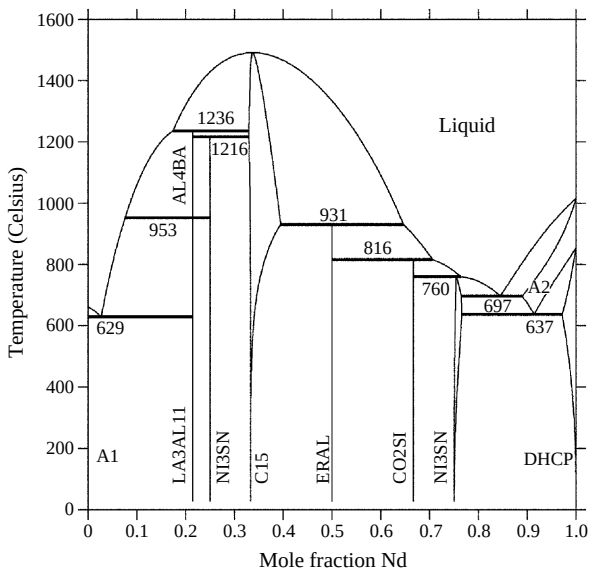

Fig. 3. The calculated Al–Nd system (slightly revised with respect to [12]).

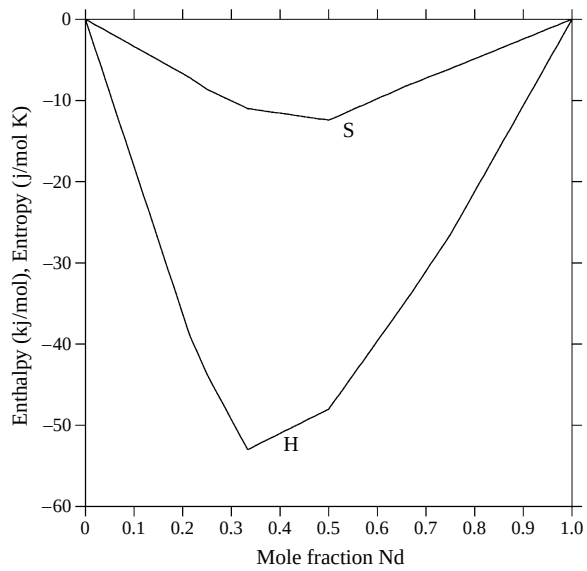

Fig. 4. Al–Nd system: thermodynamic functions of formation calculated at 25 °C.

of formation and mixing are reported in Figs. 4 and 5 respectively. It can be observed that differences between the present and the previous calculations mainly concern entropy, while enthalpies are almost unaffected.

5.4. Optimization of the Al–Ce–Nd ternary system

The calculated isothermal section at 500 °C is shown in Fig. 6. In the same figure the invariant phase compositions experimentally determined by [15] are also shown for comparison: a good agreement may be noted. Experimentally determined tie-lines reported in [15] are also in good agreement with the calculated ones. Phase equilibria in the regions with $x(\mathrm{Al}) < 0.25$ and $x(\mathrm{Al}) > 0.67$, not investigated by [15], have been predicted in this work. Only for a few phases a ternary parameter was needed in order to well reproduce the experimental phase equilibria.

The vertical section at $x(\mathrm{Ce}) = x(\mathrm{Nd})$ is shown in Fig. 7. Solid–liquid equilibria have not been studied systematically. Nevertheless, in order to evaluate a reasonable ternary interaction parameter for the liquid phase, liquidus temperatures at three key compositions were determined by DTA (see [15]) and used in the optimization.

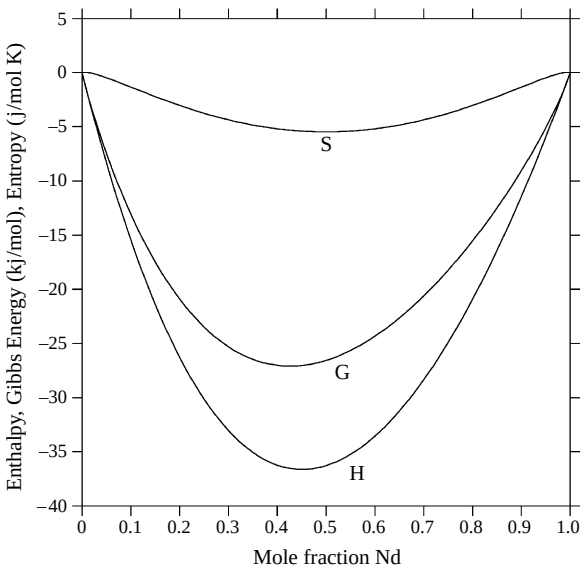

Fig. 5. Al–Nd system: thermodynamic functions of mixing calculated at 1500 °C.

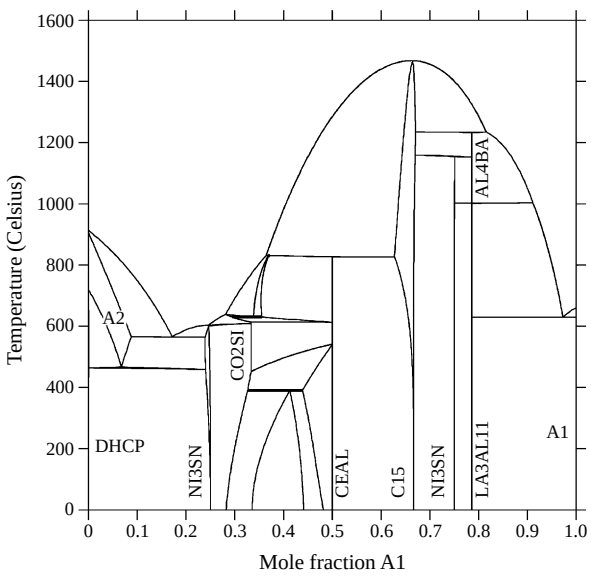

Fig. 7. Al–Ce–Nd system: vertical section calculated at $x(\mathrm{Ce}) = x(\mathrm{Nd})$.

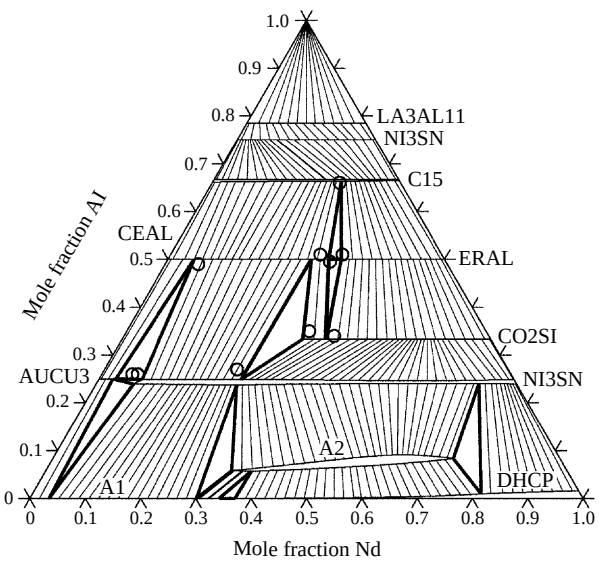

Fig. 6. Al–Ce–Nd system: isothermal section calculated at 500 °C. Open circles represent the experimentally determined invariant phase compositions.

In our opinion the resulting positive interaction parameter has to be related to the use of the Muggianu interpolation formula [3] instead of the Toop one [2] (not implemented in the software used for the optimization), which would be more appropriate in this case where two of the three component elements (Ce and Nd) are very similar to each other and quite different from the third one. In Al–Ce–Nd indeed, where we have comparable strong interactions in Ce–Al and Nd–Al subsystems, and substantially ideal mixing in Ce–Nd, the Muggianu interpolation formula tends to over-estimate the liquid stability in the Ce–Nd-rich region of the system. Then the positive ternary interaction terms partially compensate for the inadequacy of the interpolation formula. It may be finally observed that the use of the Kohler interpolation formula [1] would reduce the above-mentioned effect, but would still require some correction. For this reason and for consistency with a number of other systems already optimized, and considering the very partial experimental information available, it has been decided to keep the Muggianu interpolation and to introduce just one ternary interaction parameter.

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Appendix

The functions *GHSERXX* are the Gibbs energy functions of the pure elements in the SER state (A1 for Al and Ce, DHCP for Nd). These and the other functions *GYYYXX*, representing the Gibbs energy functions of the pure elements in different phases, have been taken from [5] and are not explicitly reported in this paper.

Parameter values are expressed in J mol^{-1} (the mole being defined according to the phase model and formula). Underlined coefficients are those optimized in this work, Al–Ce coefficients are taken from [12].

LIQUID Sublattice model: (Al, Ce, Nd)

G(LIQUID, Al; 0)-H^{298}(A1, Al; 0) = +GLIQAL
G(LIQUID, Ce; 0)-H^{298}(A1, Ce; 0) = +GLIQCE
G(LIQUID, Nd; 0)-H^{298}(DHCP, Nd; 0) = +GLIQND
L(LIQUID, Al, Ce; 0) = $-141\,000 + 43.82 * T$
L(LIQUID, Al, Ce; 1) = $-55\,600 + 23.68 * T$
L(LIQUID, Al, Ce; 2) = -1686
L(LIQUID, Al, Nd; 0) = $-\underline{145\,100} + \underline{44.97} * T$
L(LIQUID, Al, Nd; 1) = $-\underline{28\,030}$
L(LIQUID, Al, Nd; 2) = $-\underline{6708}$
L(LIQUID, Al, Ce, Nd; 0) = $+\underline{32\,980}$

A1 Sublattice model: $(Al, Ce, Nd)_1(Va)_1$

G(A1, Al:Va; 0)-H^{298}(A1, Al; 0) = +GHSERAL
G(A1, Ce:Va; 0)-H^{298}(A1, Ce; 0) = +GHSERCE
G(A1, Nd:Va; 0)-H^{298}(DHCP, Nd; 0) = +GFCCND

A2 Sublattice model: $(Al, Ce, Nd)_1(Va)_3$

G(A2, Al:Va; 0)-H^{298}(A1, Al; 0) = $+10\,083 - 4.813 * T$ + GHSERAL
G(A2, Ce:Va; 0)-H^{298}(A1, Ce; 0) = +GBCCCE
G(A2, Nd:Va; 0)-H^{298}(DHCP, Nd; 0) = +GBCCND
L(A2, Al, Ce:Va; 0) = $-100\,000 + 42.1 * T$
L(A2, Al, Nd:Va; 0) = $-\underline{136\,060} + \underline{30.43} * T$
L(A2, Al, Nd:Va; 1) = $-\underline{35\,970}$

DHCP Sublattice model: (Al, Ce, Nd)

G(DHCP, Al; 0)-H^{298}(A1, Al; 0) = $+5781 - 1.8 * T$ + GHSERAL
G(DHCP, Ce; 0)-H^{298}(A1, Ce; 0) = $-161 + .568\,86 * T$ + GHSERCE
G(DHCP, Nd; 0)-H^{298}(DHCP, Nd; 0) = +GHSERND
L(DHCP, Al, Nd; 0) = $-\underline{132\,800} + \underline{33.06} * T$
L(DHCP, Al, Nd; 1) = $-\underline{43\,040}$

C15 Sublattice model: $(Al, Ce, Nd)_1(Al, Ce, Nd)_2$

G(C15, Al:Al; 0)-3H^{298}(A1, Al; 0) = $+3 *$ GLAVAL
G(C15, Ce:Ce; 0)-3H^{298}(A1, Ce; 0) = $+3 *$ GLAVCE
G(C15, Nd:Nd; 0)-3H^{298}(DHCP, Nd; 0) = $+3 *$ GLAVND
G(C15, Ce:Al; 0)-2H^{298}(A1, Al; 0)-H^{298}(A1, Ce; 0) = $-150\,000 + 30 * T + 2 *$ GHSERAL + GHSERCE
G(C15, Al:Ce; 0)-H^{298}(A1, Al; 0)-2H^{298}(A1, Ce; 0) = $+180\,000 - 30 * T$ + GHSERAL + $2 *$ GHSERCE
G(C15, Nd:Al; 0)-2H^{298}(A1, Al; 0)-H^{298}(DHCP, Nd; 0) = $-\underline{159\,000} + \underline{33} * T + 2 *$ GHSERAL + GHSERND
G(C15, Al:Nd; 0)-H^{298}(A1, Al; 0)-2H^{298}(DHCP, Nd; 0) = $+\underline{189\,000} - \underline{33} * T$ + GHSERAL + $2 *$ GHSERND
G(C15, Ce:Nd; 0)-H^{298}(A1, Ce; 0)-2H^{298}(DHCP, Nd; 0) = +GLAVCE + $2 *$ GLAVND
G(C15, Nd:Ce; 0)-2H^{298}(A1, Ce; 0)-H^{298}(DHCP, Nd; 0) = +GLAVND + $2 *$ GLAVCE
L(C15, Ce, Nd:Al; 1) = $+\underline{5000}$

NI3SN Sublattice model: $(Al, Ce, Nd)_3(Al, Ce, Nd)_1$

G(NI3SN, Al:Al; 0)-4H^{298}(A1, Al; 0) = $+\underline{20\,000} + 4 *$ GHSERAL
G(NI3SN, Ce:Ce; 0)-4H^{298}(A1, Ce; 0) = $+\underline{20\,000} + 4 *$ GHSERCE
G(NI3SN, Nd:Nd; 0)-4H^{298}(DHCP, Nd; 0) = $+\underline{20\,000} + 4 *$ GHSERND
G(NI3SN, Ce:Al; 0)-H^{298}(A1, Al; 0)-3H^{298}(A1, Ce; 0) = $-101\,000 + 28.66 * T$ + GHSERAL + $3 *$ GHSERCE
G(NI3SN, Al:Ce; 0)-3H^{298}(A1, Al; 0)-H^{298}(A1, Ce; 0) = $-180\,000 + 41.28 * T + 3 *$ GHSERAL + GHSERCE
G(NI3SN, Nd:Al; 0)-H^{298}(A1, Al; 0)-3H^{298}(DHCP, Nd; 0) = $-\underline{110\,000} + \underline{28.17} * T$ + GHSERAL + $3 *$ GHSERND
G(NI3SN, Al:Nd; 0)-3H^{298}(A1, Al; 0)-H^{298}(DHCP, Nd; 0) = $-\underline{175\,000} + \underline{34.55} * T + 3 *$ GHSERAL + GHSERND
G(NI3SN, Ce:Nd; 0)-3H^{298}(A1, Ce; 0)-H^{298}(DHCP, Nd; 0) = $+20\,000 + 3 *$ GHSERCE + GHSERND
G(NI3SN, Nd:Ce; 0)-H^{298}(A1, Ce; 0)-3H^{298}(DHCP, Nd; 0) = $+20\,000 + 3 *$ GHSERND + GHSERCE
L(NI3SN, Ce, Nd:AL; 1) = $-\underline{20\,000}$

CEAL Sublattice model: $(Ce, Nd)_1(Al)_1$

G(CEAL, Ce:Al; 0)-H^{298}(A1, Al; 0)-H^{298}(A1, Ce; 0) = $-86\,000 + 19.07 * T$ + GHSERAL + GHSERCE
G(CEAL, Nd:Al; 0)-H^{298}(A1, Al; 0)-H^{298}(DHCP, Nd; 0) = $-\underline{95\,540} + \underline{24.87} * T$ + GHSERAL + GHSERND

ERAL Sublattice model: $(Ce, Nd)_1(Al)_1$

G(ERAL, Ce:Al; 0)-H^{298}(A1, Al; 0)-H^{298}(A1, Ce; 0) = $-\underline{85\,300} + \underline{19.07} * T$ + GHSERAL + GHSERCE
G(ERAL, Nd:Al; 0)-H^{298}(A1, Al; 0)-H^{298}(DHCP, Nd; 0) = $-\underline{96\,000} + \underline{24.87} * T$ + GHSERAL + GHSERND

CO2SI Sublattice model: $(Ce, Nd)_2(Al)_1$

$G(CO2SI, Ce{:}Al; 0)\text{-}H^{298}(A1, Al; 0)\text{-}2H^{298}(A1, Ce; 0) = -\underline{93\,290} + \underline{24} * T + GHSERAL + 2 * GHSERCE$

$G(CO2SI, Nd{:}Al; 0)\text{-}H^{298}(A1, Al; 0)\text{-}2H^{298}(DHCP, Nd; 0) = -\underline{102\,000} + \underline{24.1} * T + GHSERAL + 2 * GHSERND$

LA3AL11 Sublattice model: $(Ce, Nd)_3(Al)_{11}$

$G(LA3AL11, Ce{:}Al; 0)\text{-}11H^{298}(A1, Al; 0)\text{-}3H^{298}(A1, Ce; 0) = -581\,000 + 135.6 * T + 11 * GHSERAL + 3 * GHSERCE$

$G(LA3AL11, Nd{:}Al; 0)\text{-}11H^{298}(A1, Al; 0)\text{-}3H^{298}(DHCP, Nd; 0) = -\underline{545\,000} + \underline{100.5} * T + 11 * GHSERAL + 3 * GHSERND$

AL4BA Sublattice model: $(Ce, Nd)_3(Al)_{11}$

$G(AL4BA, Ce{:}Al; 0)\text{-}11H^{298}(A1, Al; 0)\text{-}3H^{298}(A1, Ce; 0) = -541\,000 + 104.6 * T + 11 * GHSERAL + 3 * GHSERCE$

$G(AL4BA, Nd{:}Al; 0)\text{-}11H^{298}(A1, Al; 0)\text{-}3H^{298}(DHCP, Nd; 0) = -\underline{533\,000} + \underline{90.71} * T + 11 * GHSERAL + 3 * GHSERND$

AUCU3 Sublattice model: $(Al)_1(Ce, Nd)_3$

$G(AUCU3, Al{:}Ce; 0)\text{-}H^{298}(A1, Al; 0)\text{-}3H^{298}(A1, Ce; 0) = -97\,000 + 21.01 * T + GHSERAL + 3 * GHSERCE$

$G(AUCU3, Al{:}Nd; 0)\text{-}H^{298}(A1, Al; 0)\text{-}3H^{298}(DHCP, Nd; 0) = -100\,000 + 28.17 * T + GHSERAL + 3 * GHSERND$

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